

SOFTWARE DESIGN DESCRIPTION

StockWise

“Everyone starts somewhere.”



CPSC 490

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INTRODUCTION

1.1 Purpose

Stockwise is designed to assist new investors by providing simplified financial data, educational resources, and research tools that make investing easier to understand and more accessible.

The platform helps users analyze stocks, view key market data, and stay up to date on financial news. By presenting complex financial information in a clear, user-friendly way, Stockwise helps beginners learn about investing, make informed decisions, and build confidence in managing their investments.

1.2 Scope

Stockwise provides a centralized platform where users can explore financial information, search companies, and learn about stock performance. The system allows users to register accounts, select investment interests, search for companies, and analyze stock performance, including risk and potential return on investment (ROI).

1.3 Overview

The remaining sections will cover the basic, performance, and detailed source. system and working requirements for developing a melanoma detection tool.

- Section 2 covers the functional requirements of Stockwise, stating the user's perspective and its major features.
- Section 3 represents a model for the system architecture for the model to be built. It also shows a brief description of the use case, sequence, data flow, and class diagrams of the model.

Functional Requirements

1. The application displays a welcome screen.

Description: When the user opens the application, a welcome screen is displayed, providing options to either log in to an existing account or create a new account.

Precondition: The user launches the application.

Postcondition: The welcome screen with login and registration options is displayed.

2. The application allows new users to register.

Description: The application allows first-time users to create an account by entering a username, email address, and password.

Precondition: The user selects the registration option from the welcome screen.

Postcondition: The user account is successfully created and stored in the system.

3. The application allows existing users to log in.

Description: Registered users can log in to the application using their valid username or email and password.

Precondition: The user must already have a registered account.

Postcondition: The user successfully accesses their account and is directed to the homepage dashboard.

4. The application displays a homepage dashboard.

Description: After logging in, the application displays a homepage dashboard that provides recommended financial articles and company insights based on available market data.

Precondition: The user must be logged in to the application.

Postcondition: The homepage dashboard with financial recommendations and insights is displayed.

5. The application allows users to select interests.

Description: The application allows users to select preferred sectors such as Artificial Intelligence, Technology, or Healthcare to personalize the information and recommendations they receive.

Precondition: The user must be logged in and accessing the dashboard.

Postcondition: The selected interests are saved and used to customize future recommendations.

6. The application recommends articles.

Description: The system analyzes financial news sources and filters articles to display those most relevant to the user's selected interests.

Precondition: The user has selected their areas of interest.

Postcondition: Personalized financial articles are displayed on the dashboard.

7. The application allows company search.

Description: The application provides a search feature that allows users to search for specific companies or market sectors.

Precondition: The user must be logged in and on the dashboard.

Postcondition: The system displays relevant results for the searched company or sector.

8. The application displays stock performance.

Description: The system retrieves and displays stock performance information, including price history, trends, and other market data for selected companies.

Precondition: The user searches for or selects a company.

Postcondition: The stock performance data for the selected company is displayed.

9. The application calculates risk and ROI.

Description: The system analyzes market data and stock performance to estimate potential investment risk and expected return on investment (ROI).

Precondition: The user views stock or investment data within the application.

Postcondition: The calculated risk level and ROI estimation are displayed to the user.

10. The application updates recommendations.

Description: The system continuously updates financial articles and company recommendations based on the user's searches, selected interests, and activity within the application.

Precondition: The user interacts with the application through searches or content engagement.

Postcondition: Updated personalized recommendations are displayed on the dashboard.

11. The application allows users to create a watchlist.

Description: Users can add selected companies or stocks to a personal watchlist for easier tracking.

Precondition: The user is logged in and viewing a company or stock page.

Postcondition: The selected stock is added to the user's watchlist.

12. The application allows users to remove stocks from the watchlist.

Description: Users can remove previously saved stocks from their watchlist.

Precondition: The user is logged in and accessing their watchlist.

Postcondition: The selected stock is removed from the watchlist.

13. The application displays the watchlist.

Description: The system displays all stocks saved in the user's watchlist along with their latest prices and performance.

Precondition: The user must be logged in.

Postcondition: The watchlist with current stock information is displayed.

14. The application provides stock price alerts.

Description: The system allows users to set price alerts for stocks in their watchlist.

Precondition: The user selects a stock and sets a price alert condition.

Postcondition: The system stores the alert and notifies the user when the condition is met.

15. The application allows users to view historical stock charts.

Description: The system displays historical stock price charts over different time periods, such as 1 month, 6 months, or 1 year.

Precondition: The user selects a company or stock.

Postcondition: A historical price chart is displayed.

16. The application allows users to view company information.

Description: The system provides detailed information about companies, including industry, market cap, and financial metrics.

Precondition: The user selects a company from search results or recommendations.

Postcondition: The company information page is displayed.

17. The application allows users to save educational content.

Description: Users can save articles or educational resources for later viewing.

Precondition: The user is logged in and viewing an article.

Postcondition: The article is saved to the user's saved content list.

18. The application allows users to update account information.

Description: Users can update their account details, such as username, email, or password.

Precondition: The user is logged in and accesses account settings.

Postcondition: The updated account information is saved.

19. The application allows users to log out.

Description: The application allows users to securely log out of their accounts.

Precondition: The user is currently logged in.

Postcondition: The user is logged out and returned to the welcome screen.

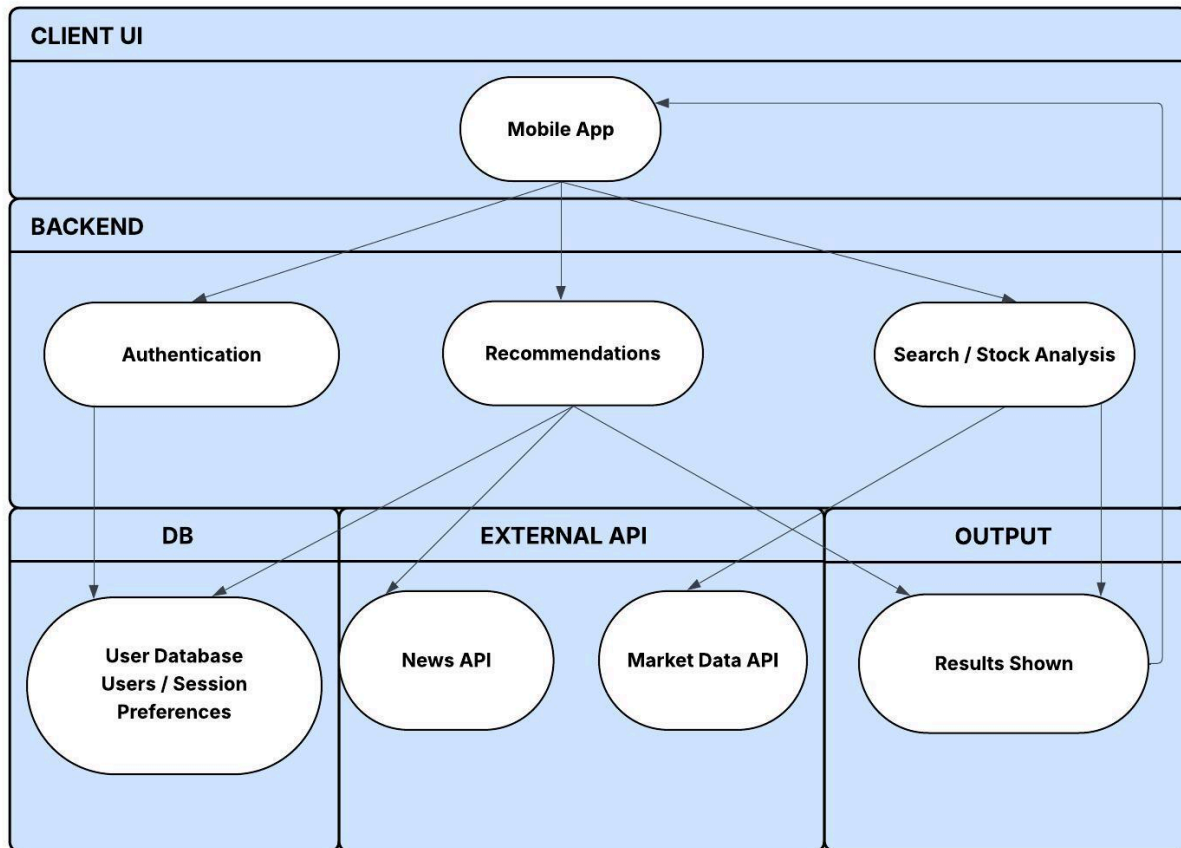
20. The application stores user data securely.

Description: The system securely stores user account data and activity information in the database.

Precondition: The user interacts with the system or creates an account.

Postcondition: User data is saved securely and protected from unauthorized access.

System Architecture



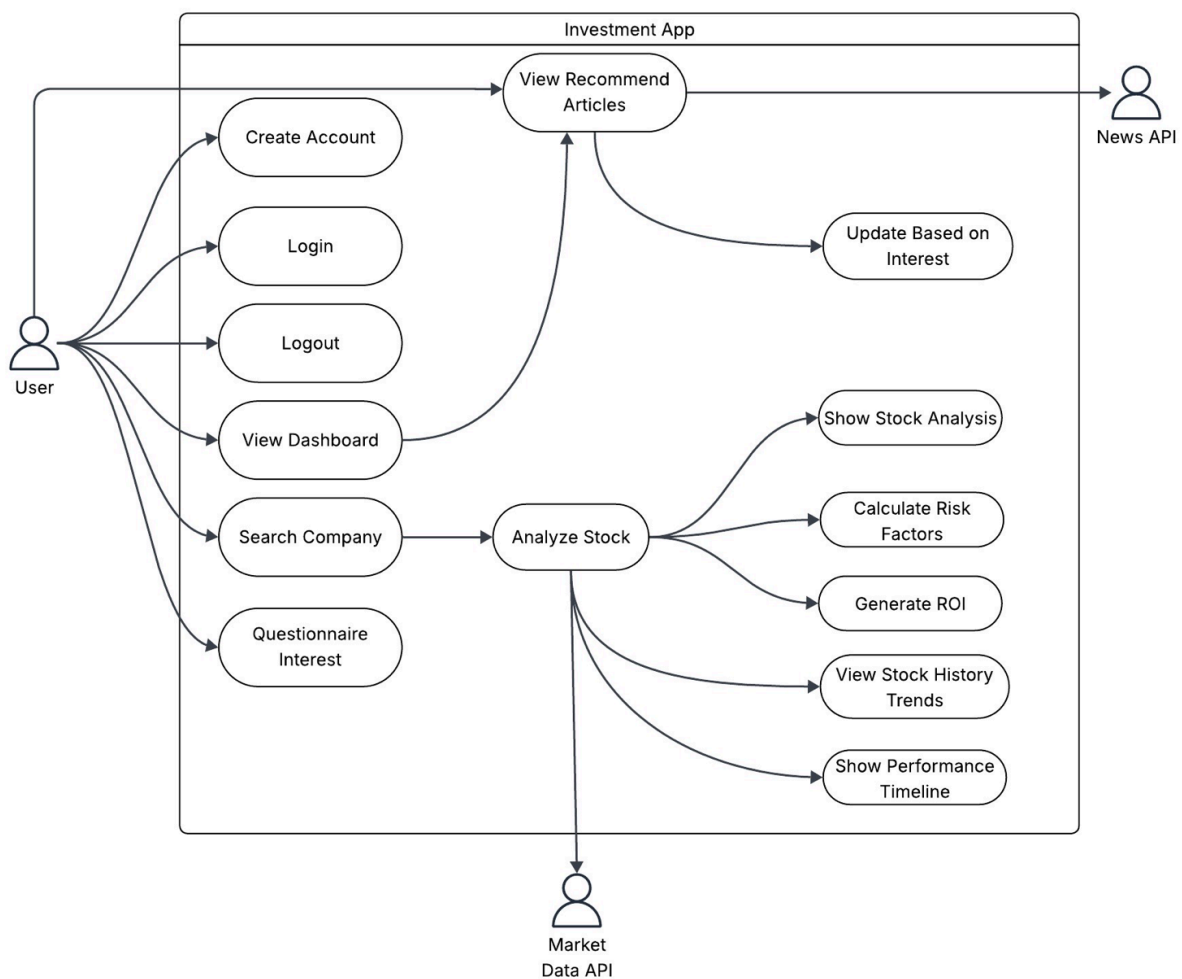
At the top is the Client UI, where users interact through a mobile app. The app sends requests to the backend, which handles the main application logic. The backend is divided into three main services: Authentication (verifies users), Recommendations (suggests stocks based on data and preferences), and Search/Stock Analysis (allows users to look up and analyze stocks).

The backend communicates with several resources. It accesses a database that stores user accounts, sessions, and preferences, and it retrieves financial information from external APIs, such as a News API for market news and a Market Data API for stock prices and financial data. After processing the information, the system sends the results back to Stockwise, where they are displayed to the user.

UML Diagrams

Use-Case diagram

The user can create an account, log in, log out, and access their dashboard to manage their activity within Stockwise. From there, Stockwise also allows users to complete an interest questionnaire so the system can personalize their experience and provide recommended financial articles. These articles are retrieved through a News API and updated based on the user's interests, while real-time stock data is gathered from a Market Data API. The user can search for companies and analyze stocks, where the system provides features such as calculating risk factors, generating potential ROI, viewing stock history trends, and displaying a performance timeline. Stockwise connects user actions, external data sources, and analytical tools to help users make more informed investment decisions.



sequence diagram

User opens Stockwise and creates an account or logs in.

Stockwise sends user credentials to the backend server for verification with the user database.

If valid, the system creates a user session.

First-time users complete an interest questionnaire to store financial preferences.

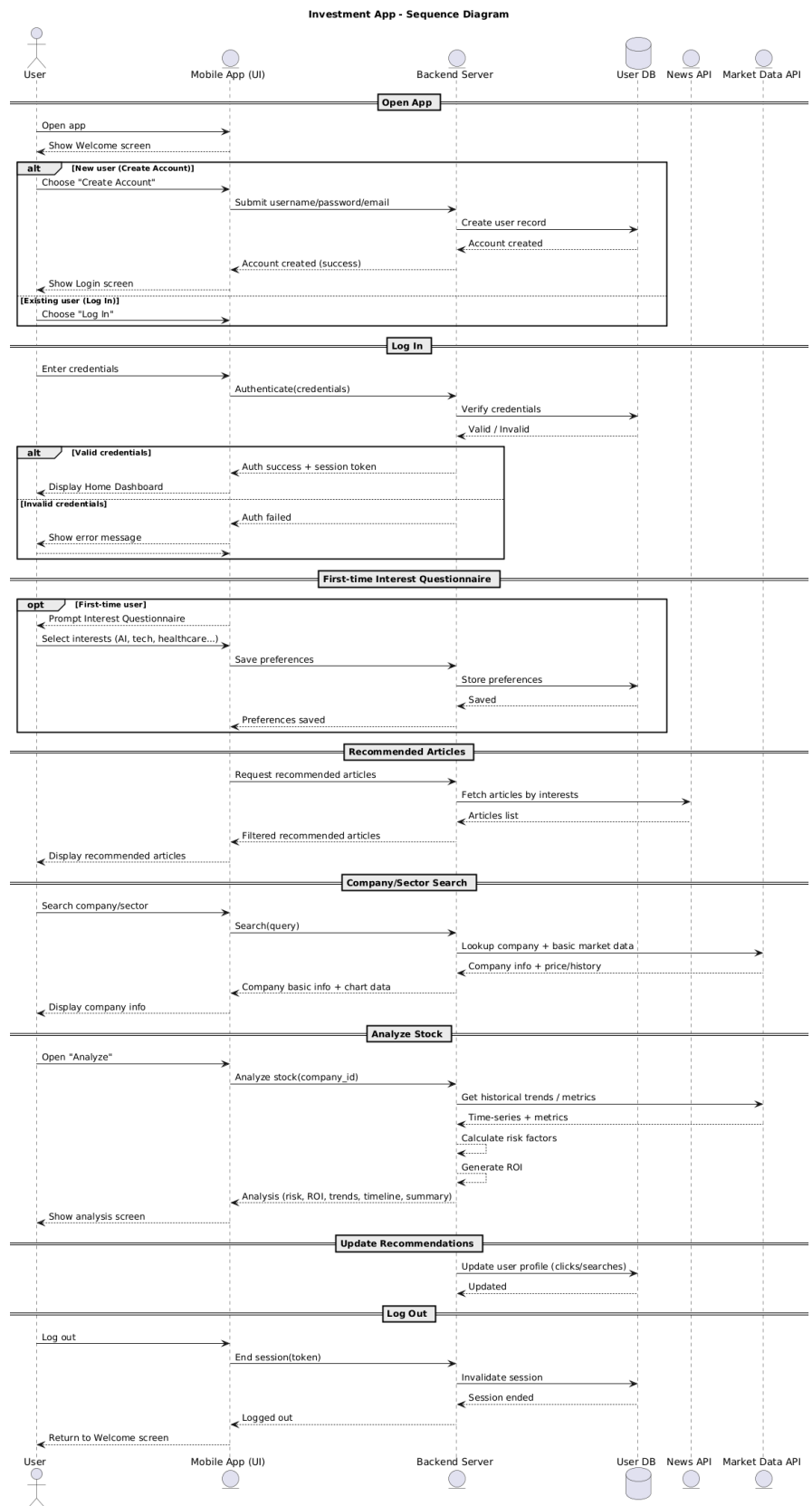
The system uses preferences to retrieve recommended financial articles from the News API.

When a user searches for a company or analyzes a stock, the backend requests data from the Market Data API.

The system calculates risk factors, potential ROI, and stock trends.

Analysis results are returned and displayed in Stockwise.

When the user logs out, the session is terminated in the database.



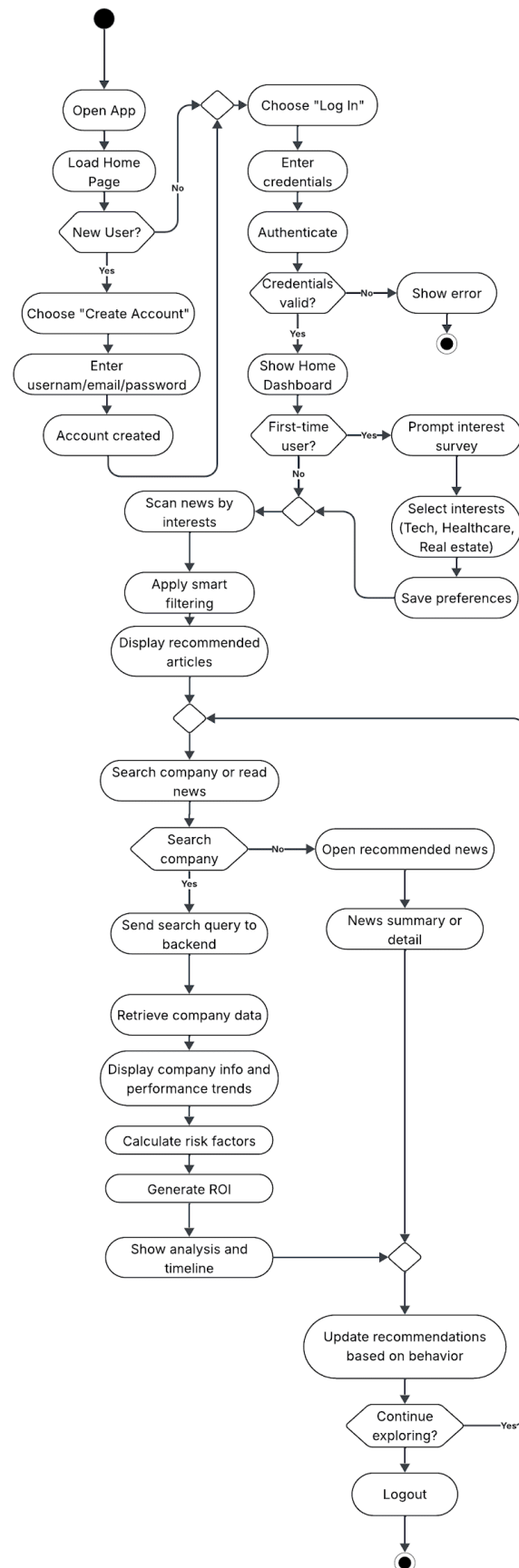
Activity Diagram

When the user opens Stockwise, the system first checks whether they are a new or returning user. New users create an account by entering their username, email, and password, while returning users log in with their credentials. If the credentials are incorrect, an error message is displayed, prompting the user to try again.

Once logged in, first-time users are guided through an interest survey, where they select topics such as technology, healthcare, or real estate. These interests are stored as personalized tags and are used to tailor the content shown to the user, including recommended financial articles and news.

From the home dashboard, users can choose to explore recommended articles based on their interests or search for a specific company. When searching for a company, the system retrieves relevant data, displays performance trends, calculates risk factors, and provides an estimated return on investment, often visualized through charts or timelines.

As users interact with the content, the system continuously updates recommendations based on their behavior and engagement, ensuring that the information remains relevant and personalized. Users can continue exploring news, stock analyses, and company data or choose to log out when they are done.



Class diagram

Users create accounts and log in through the Register and Login classes using AuthService. After logging in, users can search for companies and analyze stocks to see risk, possible returns, and market trends. Stockwise also asks users about their interests through an InterestQuestionnaire, which stores tags like technology or healthcare. These interests help the RecommendedArticles class show relevant financial articles to the user.

